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Research on brake thermal fade compensation control strategy for multi-axle electric vehicles

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ABSTRACT

Thermal fade in braking systems significantly compromises vehicle safety and stability. To address the rapid decline in braking efficiency in multi-axle electric vehicles due to thermal degradation in either motor or friction brake systems, this study investigates a five-axle vehicle equipped with a composite braking system. A thermally coupled mechanical-electric braking model is developed, incorporating both the temperature field of the brake disc and the heat dissipation of the motor brake. Furthermore, a compensation control strategy is proposed, which prioritizes regenerative braking and implements closed-loop pressure regulation of the friction brake, guided by dynamic axle load distribution and the mapped relationship among temperature, friction coefficient, and braking torque. By reconstructing the braking torque in real time, the strategy mitigates braking force attenuation under thermal fade conditions. The effectiveness of the proposed control strategy is evaluated through hardware-in-the-loop (HIL) simulations under various driving scenarios. Results demonstrate that, compared to a non-compensated strategy, braking times in the proposed method are reduced by 0.071 s, 0.220 s, and 0.490 s under normal, motor-limited, and emergency braking conditions, respectively. Meanwhile, braking distances in the proposed method are shortened by 0.964 m, 1.512 m, and 3.069 m, respectively. These findings confirm that the proposed thermal fade compensation strategy for multi-axle electric vehicles can significantly enhance braking performance stability while ensuring maximum energy recovery.

1. Introduction

The rapid development of electrification technology in the automotive industry has promoted the widespread application of various types of electric vehicles. Multi-axle electric vehicles have emerged as a critical solution for large-scale and special transportation due to their advantages, such as short energy transmission path, precise torque control, high load capacity, and low carbon emissions [1,2]. Among them, the composite braking system that integrates motor regenerative braking with traditional mechanical friction braking constitutes the core technology for such vehicles. Through the coordinated control strategy of efficient energy recovery and electromechanical braking force, it can significantly improve energy utilization and braking efficiency [3,4].

Due to the high gross vehicle weight and demanding braking conditions characteristic of multi-axle electric vehicles, brake thermal fade during frequent high-intensity braking presents a significant challenge [5,6]. Bellini C [7]. integrated experimental data from multiple brake pad materials across temperature ranges to establish an empirical correlation model between friction coefficient and temperature. Complementing this, Meng et al. [8] elucidated the formation mechanism of friction-induced hot spots in brake discs, demonstrating how thermal fields induce thermoelastic instability that degrades friction performance. Towoju et al. [9] systematically investigated how six distinct braking modes influence the thermal fade behavior of automotive braking systems, demonstrating the critical influence of braking strategy on thermal fade development. Gao et al. [10] examined the enhancement of thermal fade resistance in disc brakes through the incorporation of various fillers into resin-based, multi-fiber-reinforced friction materials. Furthermore, whereas conventional friction brakes generate heat exclusively during braking events, wheel hub motors continuously produce thermal energy during vehicle operation, resulting in significant drive-induced heat accumulation even prior to braking initiation [11]. Given that brake disc temperature elevation during vehicle deceleration adversely affects motor operating conditions, D. Tan et al.

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[12] investigated motor performance degradation under repeated aggressive acceleration cycles. Their experimental results demonstrated that after four consecutive braking cycles, winding temperatures reached 230 °C - exceeding Class H insulation limits (180 °C) and inducing severe motor torque thermal derating. Wang et al. [13] observed that continuous overloads in motors can cause permanent magnet temperatures to rise to 220 °C, resulting in demagnetization and a 93.44 % loss of torque. Additionally, A finite element model was developed by Xu et al. [14] to simulate heat transfer under braking conditions, considering thermal conduction and convection, thereby assessing how brake disc temperature rise affects the hub motor. Consequently, both conventional friction braking systems and motor braking systems can experience reduced braking efficiency due to heat accumulation under specific braking scenarios, ultimately compromising overall vehicle braking performance.

To mitigate brake thermal fade, researchers are developing active control strategies for composite braking systems. Shi et al. [15] presented an induction model with thermal and cross-coupling effects that enabled optimized MTPA and field-weakening control across the full operational range, subsequently validating its motor output predictions. To counteract transient braking issues from dynamic discrepancies and temperature-varying friction torque, Know et al. [16] developed combined concurrent control and compensation strategies. Hu [17] characterized transient thermal gradients during brake fade via finite difference modeling, thereby developing a compensation controller to mitigate the associated friction torque reduction. Han K [18] devised a novel control strategy for compensating vehicle braking torque, which accounts for the fluctuation in the frictional coefficient of brake components. Tang et al. [19] proposed a coordinated control strategy for an electro-hydraulic composite braking system, aiming to enhance both braking safety and energy recovery efficiency by leveraging braking force distribution and model predictive control. Furthermore, in the field of brake system design and control, Bravo et al. [20] optimized the parameters of hydraulic–pneumatic regenerative braking system components for both urban and highway driving cycles, thereby enabling precise brake torque control. Meng et al. [21] selected the brushless DC motor for the brake-by-wire booster. They then optimized its electromagnetic and mechanical performance via co-simulation, which resulted in high dynamic response. Zhao et al. [22] proposed an integrated control strategy for active suspension and anti-lock braking systems based on a coupled longitudinal–lateral vehicle model. Some other studies focused on the optimization of torque distribution among the axes and the development of coordinated control strategies. Qiao et al. [23] developed a driving force distribution strategy for multi-axle electric trucks that optimizes the efficiency of the electric drive system. Ghazali et al. [24] introduced a dynamic motor temperature model and proposed a temperature-adaptive torque distribution strategy. This method optimized inter-axle torque allocation by updating the efficiency map in real time, highlighting the crucial role of thermal management in energy consumption optimization. Collectively, these studies contribute to advancing the efficiency and intelligence of braking systems in electric-drive vehicles.

Based on the research, current studies on composite braking control strategies primarily focus on either thermal management and control of individual brake or motor units or coordinated control with emphasis on ride comfort. However, there remains a lack of integrated consideration of dynamic axle load transfer and its coupling effect with the real-time thermal state of the brake disc. However, in multi-axle electric vehicles, dynamic misdistribution of braking forces across axles and the coupled thermo-mechanical interactions between motors and friction brakes introduce complex thermal coupling phenomena. These complexities pose substantial challenges for thermal fade control, as existing strategies often struggle to simultaneously ensure braking safety, thermal stability, and energy recovery efficiency. To address these challenges, this study characterizes the temperature evolution and thermal rise patterns of critical components (e.g., brakes and motors), investigates the torque degradation mechanism under temperature-friction coefficient coupling, and proposes a thermal fade compensation control strategy. This strategy coordinates motor braking and friction braking based on dynamic axle load distribution to suppress and alleviate brake thermal fade, thereby enhancing braking efficiency in multi-axle electric vehicles.

2. Construction of a composite braking system and thermal model for the multi-axle electric vehicle

Traditional multi-axle vehicles typically rely on a single mechanical friction braking system, such as pneumatic, air-over-hydraulic, or fully hydraulic brakes, which do not support regenerative braking via electric motors. As a result, these vehicles exhibit limitations in braking performance, energy recovery capability, and overall energy efficiency. This study investigates an electro-mechanical composite braking system designed for multi-axle vehicles. The system integrates distributed in-wheel motor regenerative braking units into the existing air-over-hydraulic brake topology. The overall architecture for the multi-axle electric vehicle is illustrated in Fig. 1. In this configuration, the axles are mechanically linked (solid lines in Fig. 1), while the drive motors are interconnected via a high-voltage electrical bus (dashed lines in Fig. 1). Given that motor braking suffices under most driving conditions, the system prioritizes regenerative braking to enhance energy recovery.

The basic parameters of the multi-axle electric vehicle are shown in Table 1.

2.1. Modeling of air-over-hydraulic brake system

Compared to conventional vehicles, multi-axle vehicles demand superior braking performance. Fig. 2 illustrates the air-over-hydraulic brake system employed for these vehicles, comprising core components including the foot brake valve, axle control modules, air-over-hydraulic boosters and disc brakes. During the braking process, the foot brake valve receives the pedal signal. The pedal displacement is decoupled from the axle control module, so the pedal signal is transmitted to the electronic control unit (ECU). The ECU then directs control commands to individual axle control modules to regulate the air pressure at each axle. This pressure is converted into hydraulic pressure by the air-over-hydraulic booster, which actuates the caliper pistons to generate the required braking torque at the brakes. This system ensures vehicle driving safety under operational conditions.

2.1.1. Axle control module

The axle control module regulates the brake pressure at each vehicle axle. It incorporates a boost valve, a reducing valve, a backup valve, and a relay valve. It receives control instructions from the ECU through a dedicated CAN line and utilizes three solenoid valves along with a pressure sensor for closed-loop brake pressure control [25]. Fig. 3(a) shows the pressurization process, where compressed air enters the control chamber and pushes the control chamber piston downward, thereby pushing the load piston downward against spring resistance and friction. The inlet port 1 connects to the exhaust port 3, achieving pressurization. Fig. 3(c) shows the depressurization process, where intake 1 is closed, and outlet 2 and exhaust port 3 are connected, allowing compressed air to be vented to the atmosphere to achieve depressurization. Fig. 3(b) shows the pressure maintenance process, where outlet 2 is simultaneously isolated from inlet 1 and exhaust port 3 to maintain pressure.

To simplify the model, the following assumptions are adopted: (1) the thermal process of the gas inside the chamber is quasi-static; (2) the gas is treated as an ideal gas; (3) the transformation of the gas is adiabatic. Based on these assumptions and the control equations of fluid dynamics, a physical model is established to describe the dynamic behavior of each brake valve.

As shown in Fig. 3, with the initial position of the piston in (c) as x_r,

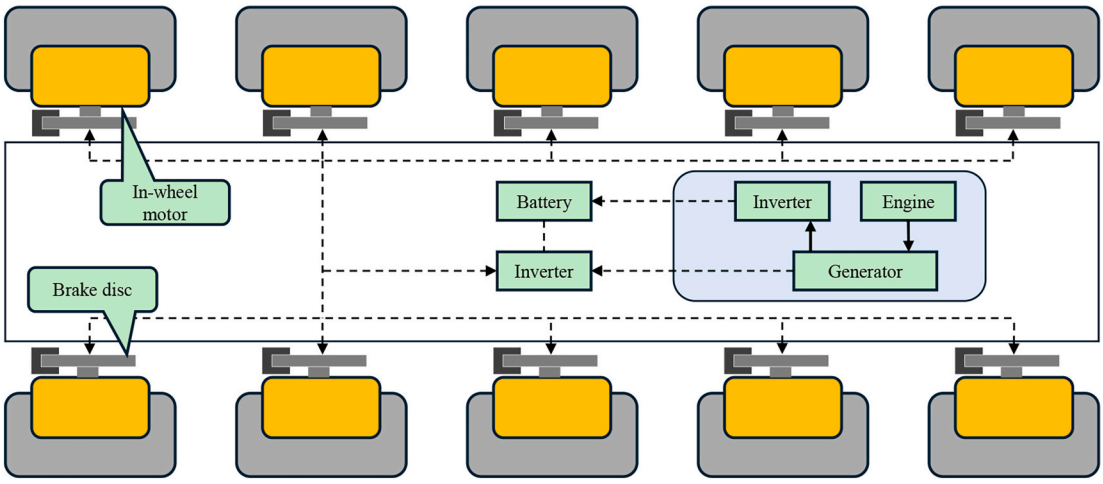


Fig. 1. The overall architecture for the multi-axle electric vehicle.

Table 1
Main structural parameters of the vehicle.

Vehicle parameters	Value	Unit
Curb weight	19000	kg
Full load mass	47000	kg
Drag coefficient	0.85	–
Windward area	7.62	m^2
Tire rolling radius	0.59	m
Rolling resistance coefficient	0.016	–
Main reduction ratio	10.81	–
Distance from each axle to 1 axle	0/3.3/6.7/9.1/11.5	m
Maximum speed	90	km/h

the middle position of the piston in (b) as x_{rm}, and the final position of the piston in (a) as x_{rt}, the motion equation of the control chamber piston in the axle control module is given by:

$$\begin{cases} m_r\ddot{x}_r = A_r p_r - A_{rl} p_c - F_f & 0 \le x_r \le x_{rm} \\ (m_r + m_{rl})\ddot{x}_r = A_r p_r - A_{rl} p_c - K_r x_r - F_0 - F_{fl} & x_{rm} \le x_r \le x_{rt} \end{cases} \quad (1)$$

Where m_r is the mass of the control piston; m_{rl} is the mass of the load piston; p_r is the air pressure entering the control chamber; p_c is the air pressure at inlet port 1; A_r is the effective area of the control piston; A_{rl} is the effective area of the relay piston; K_r is the spring stiffness of the return spring; F_0 is the prestress of the return spring; F_f is the friction between the control chamber piston and the valve body; F_{fl} is the friction between the relay piston and the valve body.

The air flow within the axle control module during braking is

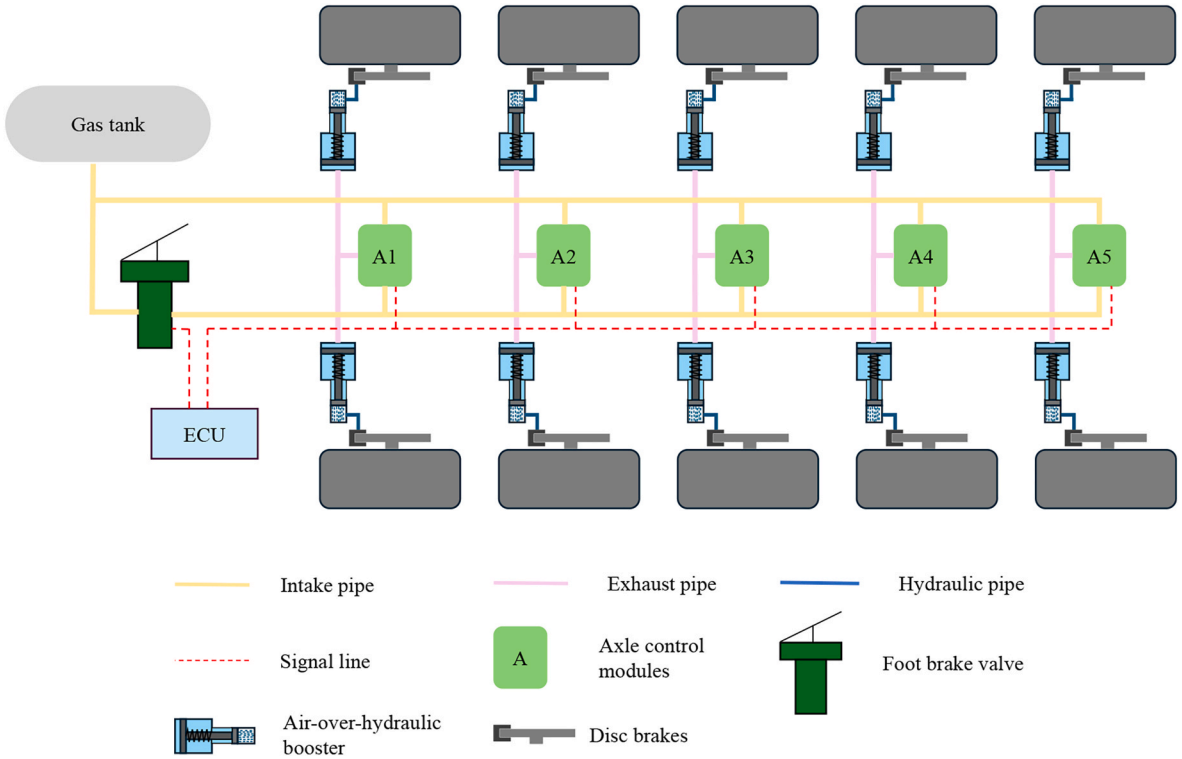


Fig. 2. Air-Over-Hydraulic brake system.

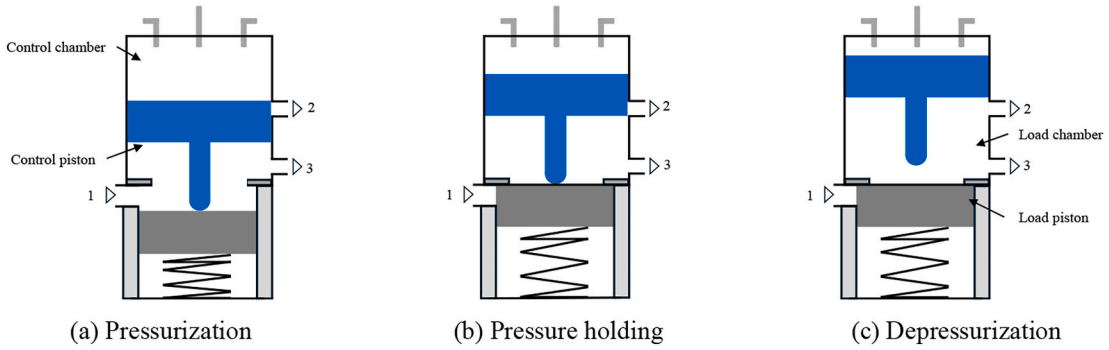


Fig. 3. Working principle of axle control module.

commonly modeled as a reversible and adiabatic process. The flow characteristic equation is:

$$Q_m = \begin{cases} C_q p_u A_d \sqrt{\dfrac{2\kappa}{RT_0(\kappa-1)}\left[\left(\dfrac{P_d}{P_u}\right)^{\frac{2}{\kappa}} - \left(\dfrac{P_d}{P_u}\right)^{\frac{\kappa+1}{\kappa}}\right]} & \dfrac{p_d}{p_u} > 0.528 \\ C_q p_u A_d \sqrt{\dfrac{2\kappa}{RT_0(\kappa+1)}}\left(\dfrac{2}{\kappa+1}\right)^{\frac{1}{\kappa-1}} & \dfrac{p_d}{p_u} \le 0.528 \end{cases} \tag{2}$$

Where C_q is the air flow coefficient; p_u is the inlet pressure; p_d is the outlet pressure; A_d is the variable throttle flow area; μ is the flow coefficient; κ is the air adiabatic index, which is 1.4; R is the air constant; T_0 is the absolute temperature.

2.1.2. Air-over-hydraulic booster

The air-over-hydraulic booster is a device that uses the area difference between the hydraulic chamber and the air chamber to increase the brake pressure. Its structure is shown in Fig. 4.

The working process of the air-over-hydraulic booster includes constant-volume adiabatic and variable-volume adiabatic processes. During braking application, compressed air rapidly pressurizes the air chamber, driving the booster piston forward in a constant-volume process. The piston movement then pressurizes the hydraulic fluid. This hydraulic pressure actuates the caliper pistons, generating friction braking force. The pressure in both chambers synchronously increases to the preset maximum, delivering peak braking torque. Upon brake release, air chamber depressurization causes the piston to retract. Hydraulic pressure decays, allowing brake fluid to return through the hydraulic circuit, and the system returns to its initial state [26].

(1) Constant volume adiabatic inflation process:

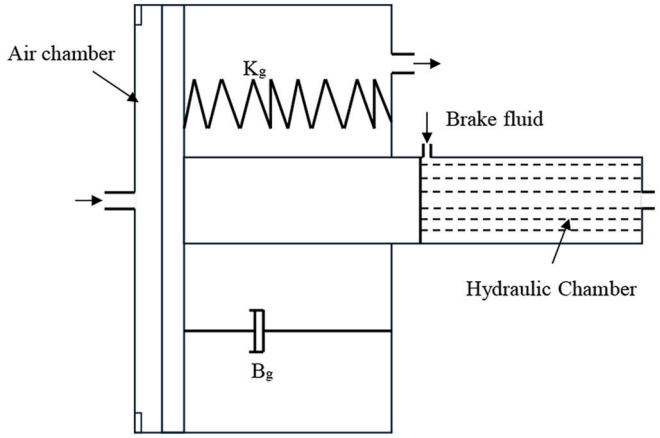


Fig. 4. Air-over-hydraulic booster.

$$t_0 = \int_{\sigma_0}^{\sigma_1} \frac{V_{g0}\,d\sigma}{\sqrt{\frac{2\kappa RT}{\kappa-1}}kA_{gen}\varphi(\sigma)} \tag{3}$$

Where $\varphi(\sigma)$ can be expressed as:

$$\varphi(\sigma) = \begin{cases} 0.2283 & 0 < \sigma \le b \\ \sqrt{\sigma^{2/\kappa} - \sigma^{(\kappa+1)/\kappa}} & \sigma > b \end{cases} \tag{4}$$

Where t_0 is the inflation time in the constant volume stage; V_{g0} is the initial volume of the air chamber; T is the absolute temperature of the compressed air, which is 293 K; A_{gen} is the effective area of the air chamber intake pipe system; σ is the absolute pressure ratio between the downstream and upstream of the air inlet; b is the critical pressure ratio of the respiratory system, with a value of 0.3.

(2) Variable volume adiabatic inflation process:

$$\frac{dp_g}{dt} = \frac{\kappa RTQ_g}{A_g(x_g + x_0)} - \frac{\kappa p_g}{x_g + x_0}\frac{dx_g}{dt} \tag{5}$$

Where p_g is the air chamber pressure; A_g is the effective area of the piston in the air chamber; x_g and x_0 are the piston displacement and starting position respectively; Q_g is the mass flow rate of the air chamber, which is given by:

$$Q_g = \frac{A_e p_e\sqrt{1-b}}{\sqrt{RT}}\omega(\sigma, b) \tag{6}$$

Where A_e is the cross-sectional area of the tracheal opening; p_e is the absolute pressure of the gas source; $\omega(\sigma, b)$ can be expressed as:

$$\omega(\sigma, b) = \begin{cases} 1 & 0 < \sigma \le \mathrm{b} \\ \sqrt{1 - (\sigma - b)^2/(1-b)^2} & \sigma > b \end{cases} \tag{7}$$

The motion equation of the booster piston subjected to air pressure is:

$$m_g\ddot{x}_g = p_g A_g - p_o A_o - K_g x_g - B_g\dot{x}_g - F_{fg} \tag{8}$$

Where m_g is the mass of the piston; x_g is the displacement of the piston; P_o is the hydraulic chamber pressure; A_o is the effective area of the pistons in the hydraulic chamber; K_g is the spring stiffness; B_g is the equivalent damping coefficient during the movement; F_{fg} is the friction force during the piston movement.

Ignoring the pipeline pressure loss and the elastic deformation of the cylinder body, the dynamic characteristic equation of the hydraulic chamber pressure is:

$$\frac{dp_o}{dt} = \frac{\beta}{V_{o0} - A_o x_g}\left(A_o\frac{dx_g}{dt} - Q_h\right) \tag{9}$$

Where β is the bulk modulus of the brake fluid; V_{o0} is the initial volume

of the hydraulic chamber; Q_h is the volume flow rate of the brake fluid out of the hydraulic chamber, which can be obtained according to Poiseuille's law [17]:

$$Q_h = \frac{\pi d^4 (p_O - p_B)}{128 \eta_b l} \tag{10}$$

Where d is the pipe diameter; p_B is the brake caliper hydraulic pressure; η_b is the kinematic viscosity of the brake fluid; l is the pipe length.

2.1.3. Disc brakes

When the vehicle brakes, the brake fluid pressure generated by the air-over-hydraulic booster increases, and the pressure input to the brake caliper hydraulic chamber also increases, forcing the brake piston to overcome the return spring force to produce a clamping force on the brake disc, thereby controlling the brake. The flow rate entering the hydraulic chamber Q_l can be expressed as:

$$Q_l = 2A_B \frac{dx_B}{dt} + \frac{V_1}{\beta_e} \frac{dp_B}{dt} \tag{11}$$

Where A_B is the piston area of the brake caliper; V_1 is the volume of the brake caliper oil inlet chamber; β_e is the elastic modulus of the hydraulic oil; x_B is the caliper piston displacement.

Ignoring the influence of the oil mass during the piston movement, the hydraulic oil squeezes the brake caliper to brake. The motion balance equation is established as:

$$m_B \frac{d^2 x_B}{dt^2} = p_B A_B - C_B \frac{dx_B}{dt} - K_B x_B - F_B \tag{12}$$

Where m_B is the mass of the piston in the brake caliper; C_B is the equivalent damping coefficient; K_B is the equivalent stiffness; F_B is the resultant force acting on the piston except the brake fluid pressure.

This study uses the disc brake. The braking torque of the disc brake is calculated as follows:

$$T_b = 2 p_B A_b \mu_b r_b \tag{13}$$

Where A_b is the braking contact area; μ_b is the braking friction coefficient; r_b is the effective braking radius.

2.2. Regenerative braking system modeling

2.2.1. Battery model

During the operation of the vehicle, the battery system provides energy under driving conditions and stores the recovered energy under braking conditions. The multi-axle vehicle studied in this study uses a lithium iron phosphate battery module, as shown in Fig. 5, which is the equivalent circuit of the battery model.

According to Kirchhoff's voltage law and Ohm's law, the relationship between battery voltage, current and power can be obtained as:

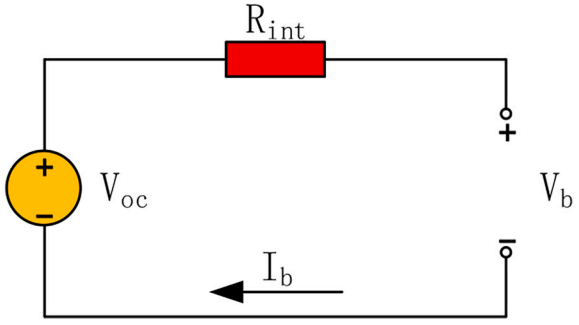


Fig. 5. Battery model equivalent circuit.

$$\begin{cases} V = V_{oc} - I_b R_{\text{int}} \\ P_b = V_b I_b = V_{oc} I_b - I_b^2 R_{int} \end{cases} \tag{14}$$

Where V is the terminal voltage; V_{oc} is the open circuit voltage; I_b is the battery current; R_{int} is the battery internal resistance; P_b is the battery power.

The discharge/charge current can be calculated as:

$$I_b = \frac{V_{oc} - \sqrt{V_{oc}^2 - 4 R_{\text{int}} P_b}}{2 R_{\text{int}}} \tag{15}$$

According to the battery charge and discharge current, the battery SOC can be estimated in real time by the ampere-hour integration method:

$$SOC(t) = SOC_0 - \frac{\int_0^t I_b dt}{Q_b} \tag{16}$$

Among them, SOC_0 is the initial value of the power battery; Q_b is the maximum capacity of the power battery.

2.2.2. Motor model

The motor converts the DC power stored in the battery into mechanical energy, providing driving force to propel the vehicle. During braking, the motor can act as a generator, converting the kinetic energy generated during braking into electrical energy, which is then stored back into the battery [27]. The motor selected for the target vehicle model in this study is a permanent magnet synchronous motor, which has the characteristics of high peak efficiency and good control stability. The motor external characteristic curve is shown in Fig. 6.

The motor output torque has certain dynamic characteristics. Therefore, this study adds a first-order inertia link to obtain the actual motor output torque:

$$T_{\text{m}} = \frac{1}{\tau s + 1} T_{m_ref} \tag{17}$$

Where T_m is the actual torque of the motor; τ is the motor response time constant; T_{m_ref} is the target braking torque of the motor.

2.3. One-dimensional thermal model of composite brake system

During the composite braking process of multi-axle electric vehicles, the high heat load generated by the motor and brake disc has a significant impact on the performance of the braking system. On the one hand, the increase in brake disc temperature may cause thermal fade; on the other hand, overheating of the motor will cause partial loss of the motor's braking function. The combined effect of the two will reduce the overall braking force of the vehicle, seriously threatening driving safety. Therefore, it is necessary to establish a mechanical-electrical-thermal coupling model of the composite brake system to achieve real-time prediction of the temperature rise of key components such as brakes and motors. The thermal simulation of the three-dimensional model involves a huge amount of computation and is not suitable for real-time use. Therefore, a one-dimensional thermal model of the braking system needs to be established.

Fig. 7 presents the thermal analysis of each module of the composite brake system. The system's thermal input originates from both frictional heat generation at the brake disc and multi-source losses in the motor, including stator core losses, winding copper losses, and eddy current losses in the permanent magnets. Heat is transferred through conduction (from the brake disc to the gearbox housing) and convective cooling via the ambient air. The thermal response of each component is determined by its heat capacity. Based on this, a one-dimensional mechanical-electrical-thermal coupling model is established as shown in Fig. 8, including a heat source calculation module, a brake thermal module (brake disc, brake pad, etc.), a motor thermal module (winding, stator,

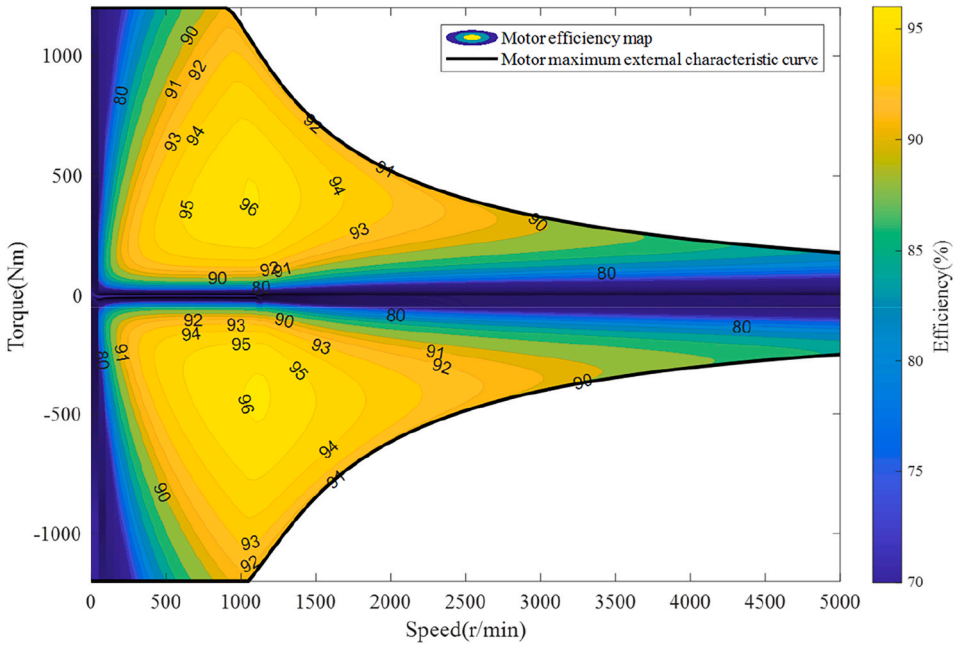


Fig. 6. Motor external characteristic curve.

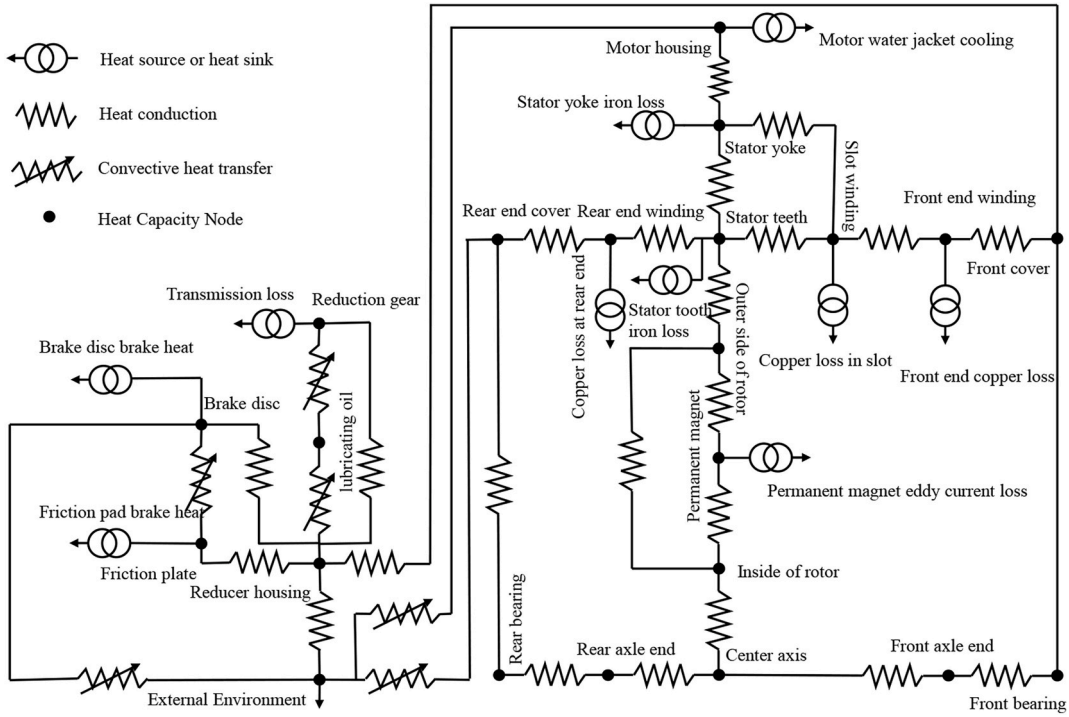


Fig. 7. Thermal analysis of each module of the composite brake system.

rotor, etc.), and a heat exchange module.

2.3.1. Brake thermal analysis model

During the braking process, the friction between the brake disc and the brake pad involves the conversion of mechanical kinetic energy into thermal energy, part of which is absorbed by the braking system, while the other part is dissipated to the surrounding environment through heat transfer. Fig. 9 shows the thermodynamic model of the brake during operation [28].

The total heat power absorbed by the brake disc is:

$$P = F_b \cdot v \cdot \eta \tag{18}$$

Where F_b is the force acting on the brake disc; v is the vehicle speed; η is the heat distribution ratio between the brake disc and the brake pad, calculated according to the following formula:

$$\eta = \frac{q_d}{q_p} = \left(\frac{k_d c_d \rho_d}{k_p c_p \rho_p}\right)^{1/2} \tag{19}$$

Where ρ_d and ρ_p are the densities of the brake disc and brake pad ma-

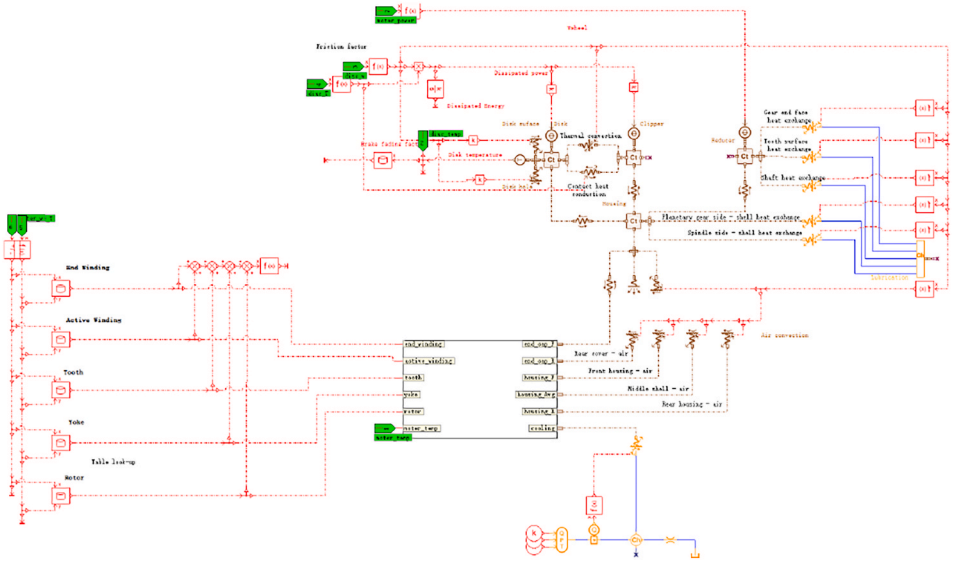


Fig. 8. One-dimensional thermal model of mechanical-electrical-thermal coupling of the composite brake system.

terials; c_d and c_p are the constant-pressure heat capacities of the brake disc and brake pad; k_d and k_p are the thermal conductivity coefficients of the brake disc and brake pad.

Considering the heat conduction, heat radiation and convection heat transfer between the surface and air of the brake disc, the heat equation of the disc brake is obtained:

$$\frac{dQ_{disc}}{dt} = P - \alpha A_{conv}(T_{disc} - T_{amb}) - \varepsilon\sigma_r A_{rad}\left(T_{disc}^4 - T_{amb}^4\right) - \frac{k_{cond}A_{cond}\left(T_{disc} - T_p\right)}{L} \tag{20}$$

The temperature of the brake disc is calculated by the above formula:

$$C_{p_disc}m_{disc}(T_{disc} - T_{amb}) = Q_{disc} \tag{21}$$

Where Q_{disc} is the heat on the brake disc; C_{p_disc} is the specific heat capacity of the brake disc; m_{disc} is the mass of the brake disc; A_{conv}, A_{rad} and A_{cond} are the heat convection area, heat radiation area and heat conduction area of the brake disc; T_{disc}, T_{amb} and T_p are the brake disc temperature, ambient temperature and brake pad temperature; α is the convection heat transfer coefficient of the brake disc surface; ε is the blackness; σ_1 is the thermal radiation constant; k_{cond} is the thermal conductivity; L is the heat conduction length.

For the brake of a certain model studied in this study, ignoring the changes of surface roughness and dimensional parameters during braking, the average convective heat transfer coefficient is only affected by the air flow characteristics of the brake, and its calculation formula is:

$$\alpha = \begin{cases} 0.04\left(\frac{k_a}{d_0}\right)\mathrm{Re}^{0.8} & Re > 2.4\times10^5 \\ 0.70\left(\frac{k_a}{d_0}\right)\mathrm{Re}^{0.55} & Re \le 2.4\times10^5 \\ Re = \omega R\rho_a d_0/\mu_a \end{cases} \tag{22}$$

Where k_a is the thermal conductivity of air; d_0 is the outer diameter of the brake disc; Re is the Reynolds number; ω is the wheel speed; ρ_a is the air density; μ_a is the air motion density.

In summary, this study models the brake disc, brake pad, housing, and reducer in the brake thermal analysis model as equivalent thermal nodes. During braking, the temperature changes of each component of the brake are detected in real time to accurately calculate the brake disc temperature for thermal fade characteristic research.

2.3.2. Motor thermal analysis model

In the braking process, apart from the thermal produced by the brake discs, the electric motor also incurs various losses, including winding copper loss, stator and rotor iron loss, permanent magnet eddy current loss, and mechanical loss. Accordingly, leveraging the key nodal structure of the motor depicted in Fig. 10, thermal resistances are incorporated to model the thermal paths between adjacent components. For instance, thermal contact conductance is applied between physically mating components like the stator core and windings. For nodes adjoining fluid channels, such as cooling water passages and the stator housing, convective thermal resistance is used. The computed losses are applied as heat sources to the corresponding nodes in the thermal network. Simulations conducted with the Motor-CAD thermal platform process these nodal inputs to generate a two-dimensional lookup table, which correlates motor losses with temperature rise. Furthermore, the motor employs a liquid cooling system for heat dissipation. During operation, the coolant circulates and absorbs heat from the motor interior through thermal exchange. Simultaneously, it releases heat to the external environment via the cooling circuit, enabling continuous temperature regulation of motor components.

The thermal capacity, thermal resistance and heat source corresponding to the above-mentioned thermal nodes are integrated to form the motor thermal network model. The thermal capacity of each node in the thermal network model can be calculated according to the basic heat transfer theory as follows:

$$C_i = \rho_i V_i c_p \tag{23}$$

Where C_i is the heat capacity of each node; ρ_i is the density of the object; V_i is the volume of the object; c_p is the specific heat capacity of the object.

The thermal resistance of the housing, stator core, rotor, magnetic steel and other components can be directly calculated using the thermal resistance formula of the hollow cylinder in heat transfer. The radial and the circumferential thermal resistance of the hollow cylinder is shown as:

$$R_{ra} = \frac{2\pi}{\phi}\frac{\ln\left(\frac{r_o}{r_i}\right)}{2\pi l_c\lambda} \tag{24}$$

$$R_{ci} = \frac{\phi}{2\lambda l_c}\frac{r_i + r_o}{r_o - r_i} \tag{25}$$

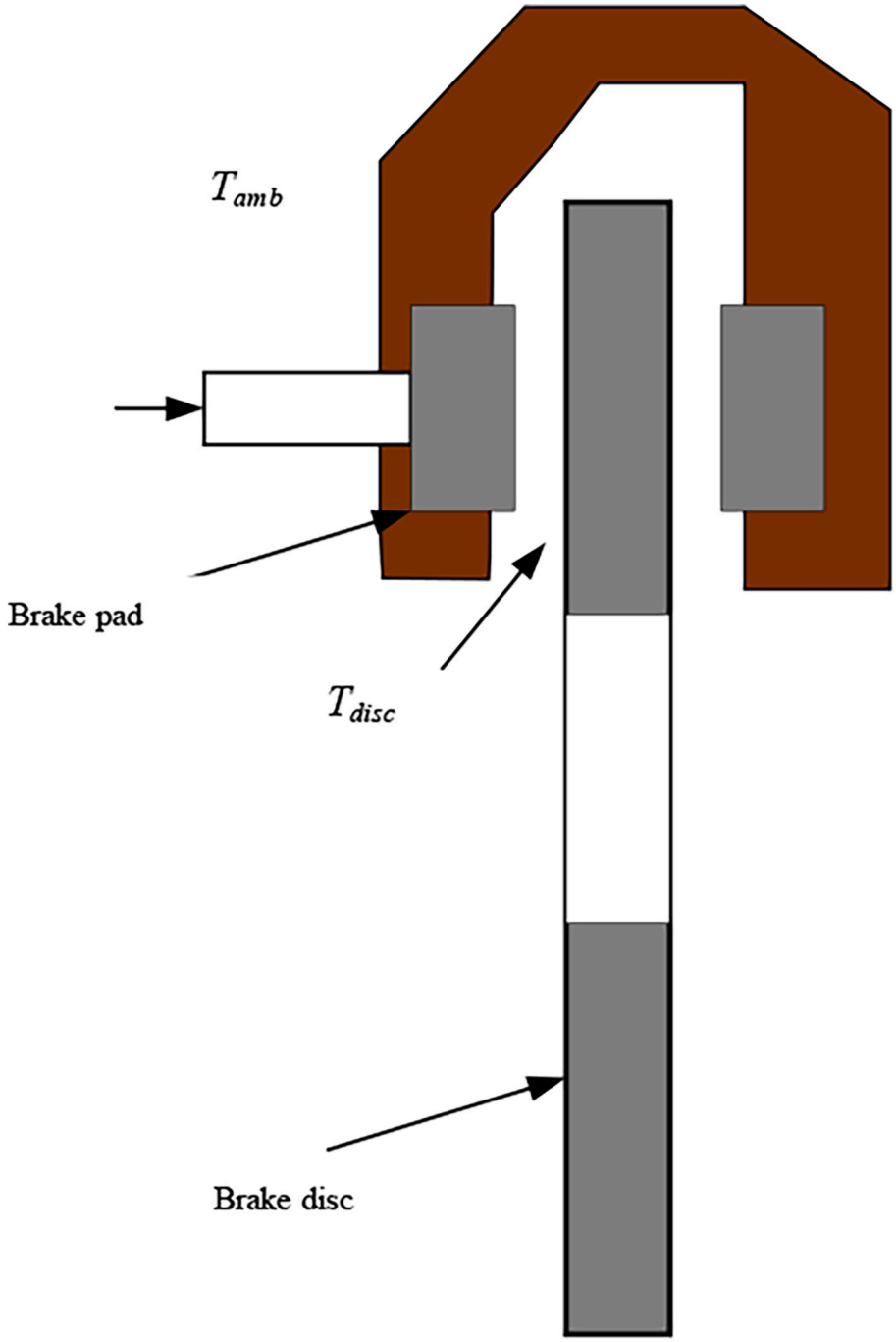


Fig. 9. Brake thermodynamic model.

Where R_{ra} is the radial thermal resistance; R_{ci} is the circumferential thermal resistance; λ is the thermal conductivity of the material; ϕ is the angle that the partial hollow cylinder occupies the complete hollow cylinder; r_i and r_o are the inner and outer diameters of the hollow cylinder respectively; l_c is the length of the hollow cylinder.

The temperature of each node changes with time. According to the law of conservation of energy, at any time, the temperature change rate of the node i and its heat capacity parameter and the adjacent node j satisfy the dynamic equilibrium relationship:

$$C_i \frac{dT_i(t)}{dt} = \sum_{i \neq j} \dot{Q}_{ij}(t) + W_i(t) \tag{26}$$

Where $W_i(t)$ is the heat source input of node i at time t; $\dot{Q}_{ij}(t)$ is the heat flow between node i connected to node j, and defines energy flowing into node i as negative and out of node i as positive. The expression is:

$$\dot{Q}_{ij} = \frac{T_i(t) - T_j(t)}{R_{ij}} \tag{27}$$

Where R_{ij} is the thermal resistance between node i and node j.

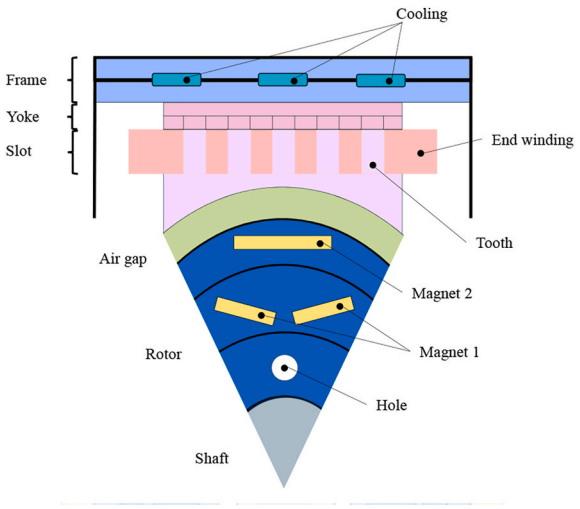


Fig. 10. Motor key node structure diagram.

3. Research on compensation control strategy of composite brake system based on brake thermal fade characteristics

3.1. Thermal fade characteristics of brakes

To investigate how brake thermal fade influences the performance of composite braking systems, we have chosen gray cast iron as the material for the brake discs, and a series of thermal fade tests are conducted using a brake performance test bench. As summarized in Table 2, the tests recorded the friction coefficient of the brake disc at various temperatures. The data collected are then fitted using the least squares method to establish a formula describing the temperature-dependent variation in the friction coefficient.

According to the scatter plot of the bench data, the quadratic polynomial is used as the basis function of the least squares fitting method, and the function of the change in the friction coefficient caused by temperature rise is obtained as follows:

$$\mu(T_B) = p_1 {T_B}^2 + p_2 T_B + p_3 \tag{28}$$

Where $\mu(T_B)$ is the sliding friction coefficient at T_B°C; p_1, p_2 and p_3 are fitting coefficients. Since the temperature span of the bench test data is large, T_B is normalized according to the mean and standard deviation, and its values are: −0.0265, −0.0437, and 0.3348. The fitting R^2 of this formula is 0.9803, indicating that the fitting function is highly consistent with the test data and can better reflect the relationship between the friction coefficient and temperature. The result of the fitting is presented in Fig. 11.

Table 2
Friction coefficient bench test data.

Serial number	Temperature(°C)	Friction coefficient
1	63.281	0.334
2	92.864	0.340
3	120.788	0.345
4	149.151	0.350
5	177.053	0.360
6	204.150	0.365
7	232.036	0.360
8	260.245	0.340
9	288.334	0.333
10	316.272	0.320
11	343.675	0.315
12	371.625	0.290
13	400.275	0.275
14	427.523	0.265
15	455.474	0.240
16	483.430	0.225
17	511.371	0.210

Based on the analysis of the test data and its fitting expression, it can be seen that there is a nonlinear attenuation relationship between the friction coefficient and the temperature. When the temperature of the brake disc is about 204 °C, the friction coefficient reaches a peak value of 0.365. After that, the temperature continues to rise, and the friction coefficient gradually decreases. When the brake temperature rises to 380 °C, the friction coefficient drops to about 0.286, which is close to the value at ambient temperature. Therefore, this study selects 380 °C as the critical threshold. When the brake disc temperature is lower than this threshold, the braking system can maintain good braking performance without additional compensation control; when the temperature is higher than this threshold, pressure compensation control of the braking system is required to ensure the safety of vehicle operation.

3.2. Compensation control of composite braking system of multi-axle electric vehicle under thermal fade condition

Informed by the thermal fade characteristics of the brake disc and the working principle of the braking system, this section formulates the compensation strategies for regenerative and friction braking. The effectiveness of the proposed control method is validated through co-simulation using Simulink and AMESim. As shown in Fig. 12, the control strategy framework centers on compensation based on motor winding and brake disc temperatures [29].

To ensure braking safety with energy recovery efficiency, this study presents a composite braking compensation control strategy. The strategy monitors the brake disc and motor winding temperatures in real time. This strategy integrates a high-precision thermodynamic model, dynamic observation of axle load, and multi-motor coordinated control. Based on considering the dynamic axle load and real-time thermal coupling of the vehicle, it distributes the braking force of each axle, and through real-time monitoring of the temperature of the brake disc and the motor windings. By setting key threshold values—such as the thermal tolerance limit of the motor insulation (180 °C) and the critical temperature for brake disc thermal fade (380 °C)—the system coordinates regenerative and friction braking compensation to ensure adequate total braking force, thereby maintaining vehicle safety. The composite brake compensation control strategy is shown in Fig. 13. The specific implementation steps are as follows.

Step 1 Initial allocation.

Axle vertical loads $F_{z,i}$ are estimated based on the driver's braking input, vehicle longitudinal acceleration, and a load transfer model. These loads are then used to proportionally assign target braking torques $T_{req,i}$ to each wheel according to the load distribution.

Step 2 Enter the control loop.

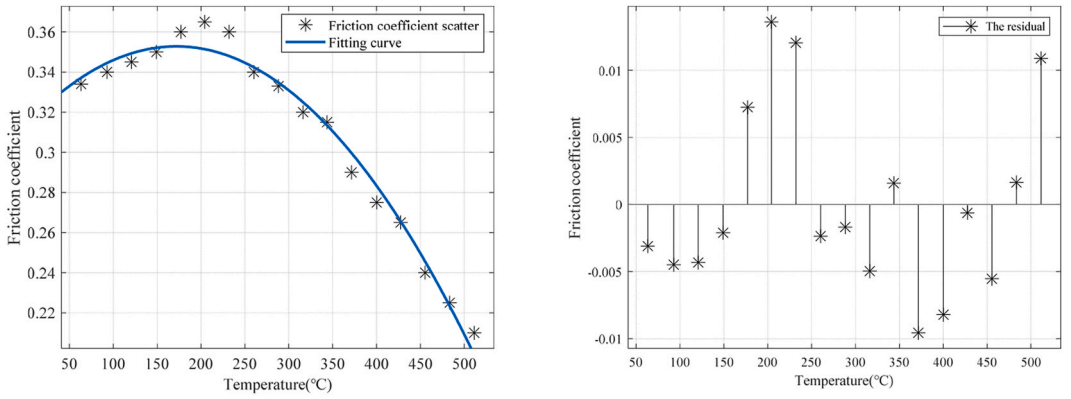
When the vehicle does not reach the target speed V_{target}, repeat the following steps (Step 2–8).

Step 3 Determine the motor working condition and conventional braking condition.

For each axle *i*.

- Check the motor winding temperature: $T_{\text{mot},i} < T_{\text{mot},thr} = 180°C$.
- Check the braking intensity: $z = a_x/g < z_{thr} = 0.7$.

If both are met, then go to "electromechanical composite braking/pure motor braking " (Step 4); otherwise go to "pure friction braking" (Step 5).



(a) Friction coefficient fitting curve (b) Friction coefficient fitting residual plot

Fig. 11. Friction coefficient bench test data fitting diagram.

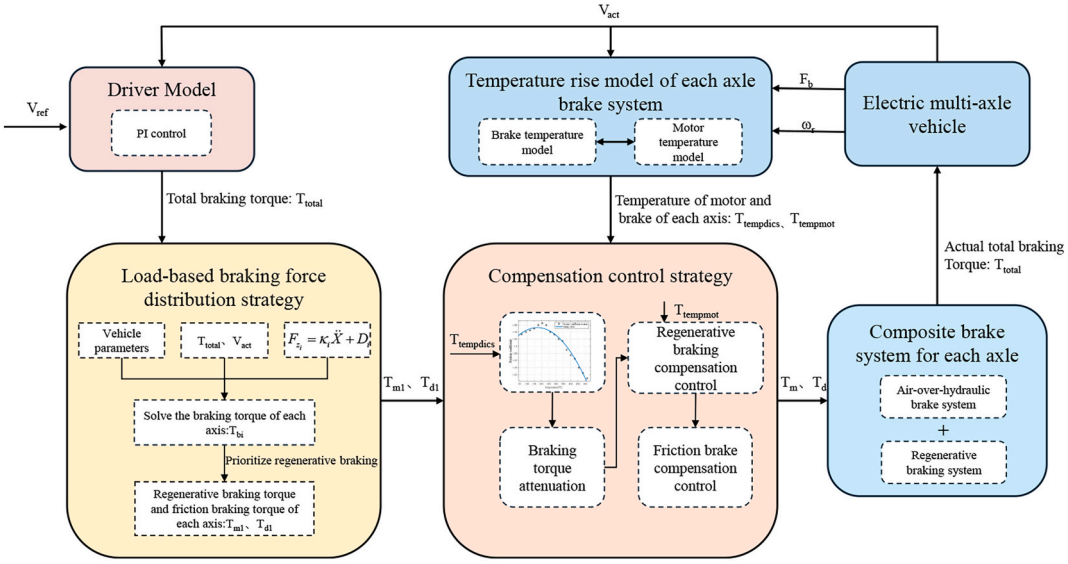


Fig. 12. Compensation control architecture of the composite braking system.

Step 4 Compare the required torque with the motor regeneration capacity.

For axle i that meets the motor winding temperature and working conditions: $T_{\mathrm{mot},i} < T_{\mathrm{mot},thr}\&z < z_{thr}$, compare target braking torques $T_{req,i}$ with the motor maximum regenerative braking capacity $T_{em,\max,i}$.

- If $T_{req,i} > T_{\mathrm{em,max},i}$, then perform "electromechanical composite braking" (Step 6);
- Otherwise, perform "pure motor braking" (Step 7).

Step 5 Pure friction braking strategy.

For axle i that do not meet the motor or working condition requirements: use pure friction braking, set the braking torque $T_{fric,i} = T_{req,i}$, and determine whether the brake disc temperature exceeds the limit: $T_{disc,i} > T_{disc,thr} = 380°C$.

- If it exceeds the limit, enter the friction brake compensation;
- Otherwise, maintain pure friction braking.

Step 6 Electromechanical composite braking strategy

For axle with $T_{req,i} > T_{\mathrm{em,max},i}$, apply motor regenerative braking torque $T_{em,i} = T_{em,\max,i}$ and friction braking torque $T_{fric,i} = T_{req,i} - T_{em,\max,i}$ at the same time, and check the temperature threshold safety condition again: $T_{\mathrm{mot},i} > T_{\mathrm{mot},thr}$ or $T_{disc,i} > T_{disc,thr}$.

- If either exceeds the limit, enter the " Regeneration/friction compensation module " (Step 8);
- Otherwise, execute directly according to the above composite strategy.

Step 7 Pure motor braking strategy

For axle with $T_{req,i} \le T_{\mathrm{em,max},i}$, apply regenerative braking $T_{em,i} = T_{req,i}$, set $T_{fric,i} = 0$, and check the motor winding temperature: $T_{\mathrm{mot},i} > T_{\mathrm{mot},thr}$.

- If it exceeds the limit, enter the " Regeneration/friction compensation module" (Step 8);
- Otherwise, execute according to pure motor braking.

Step 8 Regeneration/friction compensation module

For the axle to be compensated, the total remaining regeneration

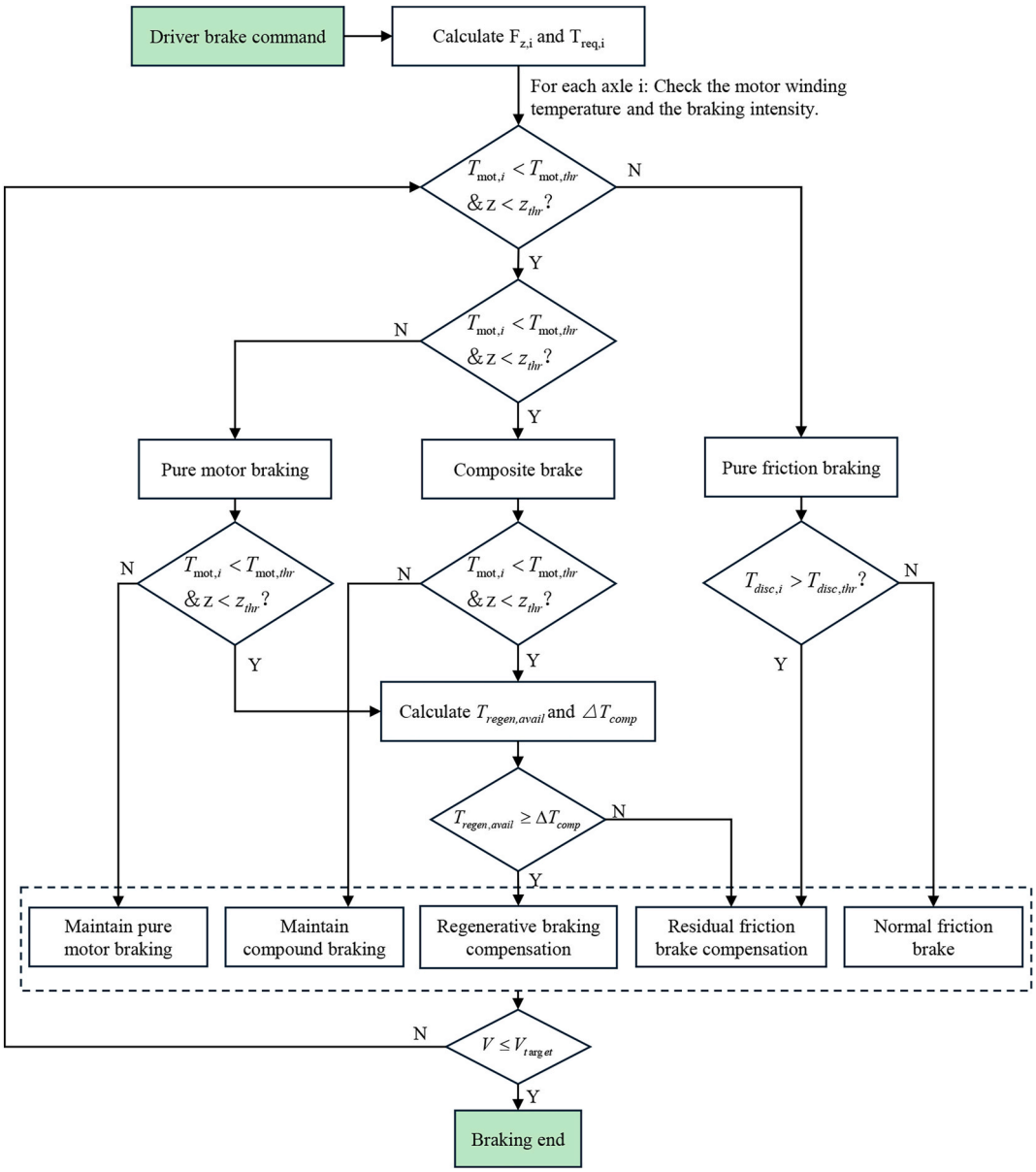


Fig. 13. Composite brake compensation control strategy.

capacity $T_{regen,avail}$ of other axles and the vehicle braking torque attenuation ΔT_{comp} caused by the temperature exceeding the threshold are counted.

- If $T_{regen,avail} \geq \Delta T_{comp}$, the other axles will perform regenerative braking compensation;
- Otherwise, the remaining compensation will be borne by friction braking.

Step 9 End judgment

If the current vehicle speed $V \leq V_{\text{target}}$, the braking process is terminated; otherwise, return to step 3 and continue the next control cycle.

3.2.1. *Design of ideal braking force distribution strategy based on axle load*

The distribution of braking force across each axle during braking is contingent upon the dynamic changes in vertical wheel load. However, a strategy that applies braking force uniformly to all axles represents an oversimplification, as it fails to account for the critical axle load transfer effect induced by braking. Therefore, when calculating and analyzing the force on a multi-axle vehicle, it is necessary to introduce the suspension deformation coordination equation, regard the suspension as an elastic element, and consider the total vertical stiffness composed of the superposition of the tire and suspension stiffness. Assuming that the vehicle body and wheels are rigid bodies, and ignoring the relative deformation between the frame and the suspension [30], the vertical load on each axle during braking is obtained as:

$$F_{z,i} = \kappa_i \ddot{X} + D_i \tag{29}$$

$$\kappa_i = \frac{m h_0 k_i \sum_{j=1}^{n} k_j (l_j - l_i)}{\sum_{j=1}^{n} \sum_{m=1}^{n} k_j k_m (l_2 - l_j)(l_m - l_j)} \tag{30}$$

$$D_i = \frac{mgk_i \sum_{j=1}^{n} k_j (l_j - l_i)(l_j - l_c)}{\sum_{j=1}^{n} \sum_{m=1}^{n} k_j k_m (l_2 - l_j)(l_m - l_j)} \tag{31}$$

Where $F_{z,i}$ is the vertical load on each axle under braking deceleration $\ddot{X}$; D_i is the static load on the i axle; κ_i is the rate of change of the vertical load on each axle to the braking deceleration, which is only related to l_i and k_i. h_g is the vehicle's center of gravity height, l_i is the distance from the first axle to each axle, and k_i is the suspension stiffness. In this study, $n = 5$.

Utilizing a high-precision braking system featuring independent single-axle control, the inter-axle braking force distribution is optimized according to the real-time dynamic load distribution ratio. The distribution relationship can be described by the following [31]:

$$\beta_i = \frac{F_{zi}}{mg} \tag{32}$$

The braking torque T_{bi} allocated to each axle is:

$$T_{bi} = T_{req} \times \beta_i \tag{33}$$

Where $i = 1,2,3,4,5$.

3.2.2. Regenerative braking compensation control

This study adopts a multi-mode coordinated compensation strategy based on dynamic axle load distribution. The regenerative braking compensation control process is shown in Fig. 14. The heat resistance threshold (H grade 180 °C) corresponding to the motor insulation level

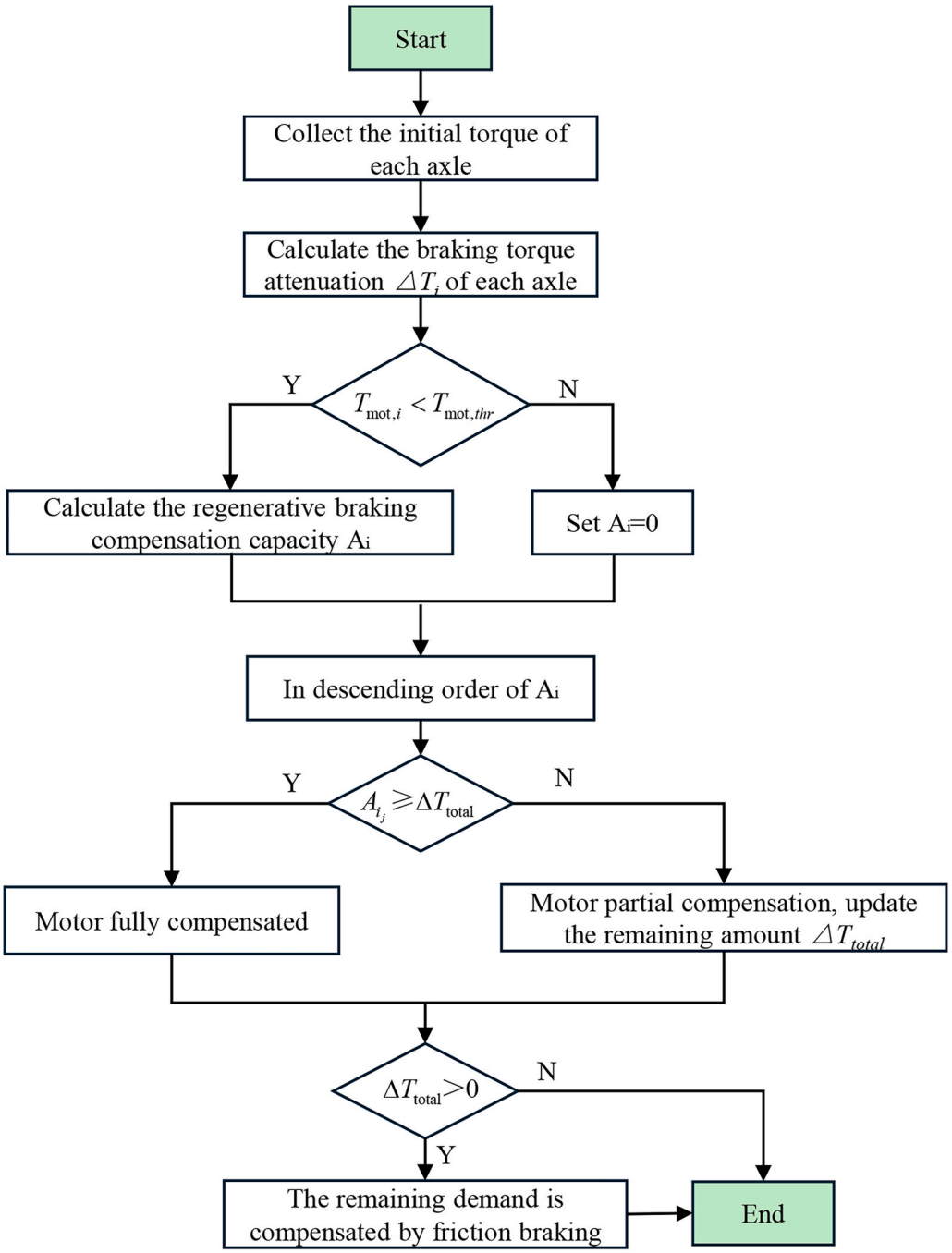


Fig. 14. Regenerative braking compensation control process.

is used as the switching basis, and the remaining capacity ranking is used as the allocation principle. The regenerative braking and friction braking are coordinated in an orderly manner according to the axle [32]. The specific steps are as follows.

Step 1 Initialization.

Set the motor temperature threshold to 180 °C and collect the initial braking torque requirements of each axle. Calculate the braking torque attenuation ΔT_i of each axle according to the thermal fade model.

Step 2 Calculate compensation capacity and sort.

Check the motor winding temperature. For the axle with motor winding temperature $T_{\mathrm{mot},i} < T_{\mathrm{mot},thr} = 180°C$, calculate its maximum available regenerative braking compensation capacity A_i; For the axle with $T_{\mathrm{mot},i} > T_{\mathrm{mot},thr}$, treat A_i as 0 (only friction braking compensation), and sort all axes by A_i from large to small to obtain the compensation priority sequence $\{A_1, A_2, A_3, A_4, A_5\}$.

Step 3 Regenerative braking compensation.

Traverse each axle i_j in sequence and let the motor with the strongest compensation capacity compensate for the global residual attenuation in turn. Calculate the total braking torque attenuation requirement of all axes: $\Delta T_{\mathrm{total}} = \sum_{i=1}^{5} \Delta T_i$.

- If $A_{i_j} \geq \Delta T_{\mathrm{total}}$, the motor of the axle fully compensates $\Delta T_{\mathrm{total}}$, and the compensation is completed.
- If $A_{i_j} < \Delta T_{\mathrm{total}}$, then the motor i compensates the maximum braking torque A_{i_j}. It can compensate, and the remaining total braking torque attenuation is updated to $\Delta T_{\mathrm{total}} \leftarrow \Delta T_{\mathrm{total}} - A_{i_j}$, and then continue to the next motor.

Step 4 Friction braking compensation.

Traverse all axes, and for the part where $\Delta T_{\mathrm{total}} > 0$ (the motor fails to fully compensate or is disabled by temperature limit), all compensation is borne by the friction braking system of the decaying axle.

Step 5 Output control instructions.

Summarize the final motor regenerative torque and friction torque of each axle to form a complete braking compensation instruction and end the process.

3.2.3. Friction brake compensation control

The friction brake compensation mode is when the motor cannot or is insufficient to provide braking force compensation, and the friction brake system bears the remaining braking force demand. Given the structural characteristics of the disc brake, the friction torque can be increased by actively increasing the wheel cylinder pressure [33,34], thereby achieving a braking effect. This study adopts the control logic shown in Fig. 15 for friction brake compensation control. The sliding mode controller takes the difference between the target braking torque and the actual braking torque as input. After the controller output is processed by the data conversion module, the corresponding control signal is generated to adjust the solenoid valve, thereby accurately controlling the output braking force. The target braking torque of each axle is obtained by real-time calculation of the compensation control strategy, while the actual braking torque is directly obtained through the air-over-hydraulic brake system model.

Set the target braking torque as $T_d(t)$, and the actual braking torque as $T(t)$. Define the error as:

$$e(t) = T_d(t) - T(t) \tag{34}$$

The designed sliding surface is:

$$s(\mathrm{t}) = \dot{e}(t) + \lambda e(t) \tag{35}$$

Where $\lambda > 0$ is the sliding surface coefficient.

In this study, considering that the intermediate variables such as the control chamber pressure and the solenoid valve core displacement in the braking system are difficult to measure, this study only uses the pressure signal to realize the closed-loop adjustment of the system, so the sliding mode control rate is designed as:

$$u(t) = -K(t) \cdot sat\left(\frac{s(t)}{\phi'}\right) \tag{36}$$

Where $sat\left(\frac{s(t)}{\phi'}\right)$ is the saturation function; $K(t)$ is the dynamically changing control gain, and satisfies the following adaptive rate:

$$\dot{K}(t) = \gamma \cdot (|s(t)| - \phi') \tag{37}$$

Where $\gamma > 0$ is the rate adjustment factor; ϕ' is the boundary layer thickness.

To verify the stability of the adaptive sliding mode control method, this study analyzes it using the Lyapunov stability theory. The following positive definite Lyapunov function is selected:

$$V(s, K) = \frac{1}{2}s^2 + \frac{1}{2\gamma}(K - K_0)^2 \tag{38}$$

Where K_0 represents the initial gain value. Taking the derivative of

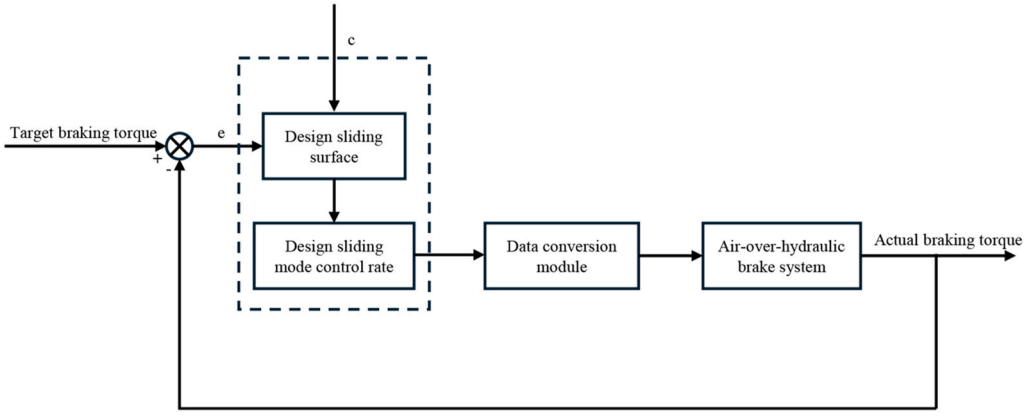


Fig. 15. Friction brake compensation control.

$V(s,K)$ yields:

$$\dot{V} = s \cdot \dot{s} + \frac{1}{\gamma}(K - K_0)\dot{K} \tag{39}$$

Substituting Equations (37) and (39) into Equation (41), the resulting expression can be approximated under ideal sliding mode conditions, neglecting the effects of the saturated boundary layer.

$$\dot{V} = -K|s| + (K - K_0)(|s| - \phi) \tag{40}$$

Since $V(s,K) = \frac{1}{2}s^2 + \frac{1}{2\gamma}(K - K_0)^2$ and $|s| > \phi$, it follows that $\dot{V} < 0$. Therefore, the system satisfies the condition for asymptotic stability.

The thermal fade compensation control strategy proposed in this paper is based on the decoupled electro-hydraulic braking system and can be implemented without interfering with the driver's pedal displacement.

4. Composite braking performance simulation results and analysis

4.1. Hardware-in-the-loop system

To evaluate the overall performance of the proposed thermal fade compensation strategy, we established a hardware-in-the-loop (HIL) simulation platform specifically designed for multi-axle vehicles with composite braking systems. The overall architecture of the platform is depicted in Fig. 16. It features a distributed structure composed of three primary components: a host computer, an xPC Target real-time simulation system, and a vehicle control unit (VCU).

The host computer executes the computational tasks for the high-fidelity vehicle dynamics model developed in the AMESim. This model captures the vehicle's longitudinal dynamics, the thermodynamic behavior of the braking system, and key components, including the pneumatic braking circuit. Following compilation, the model parameters are downloaded via high-speed Ethernet to the xPC Target real-time target machine. Equipped with a real-time kernel, this machine is

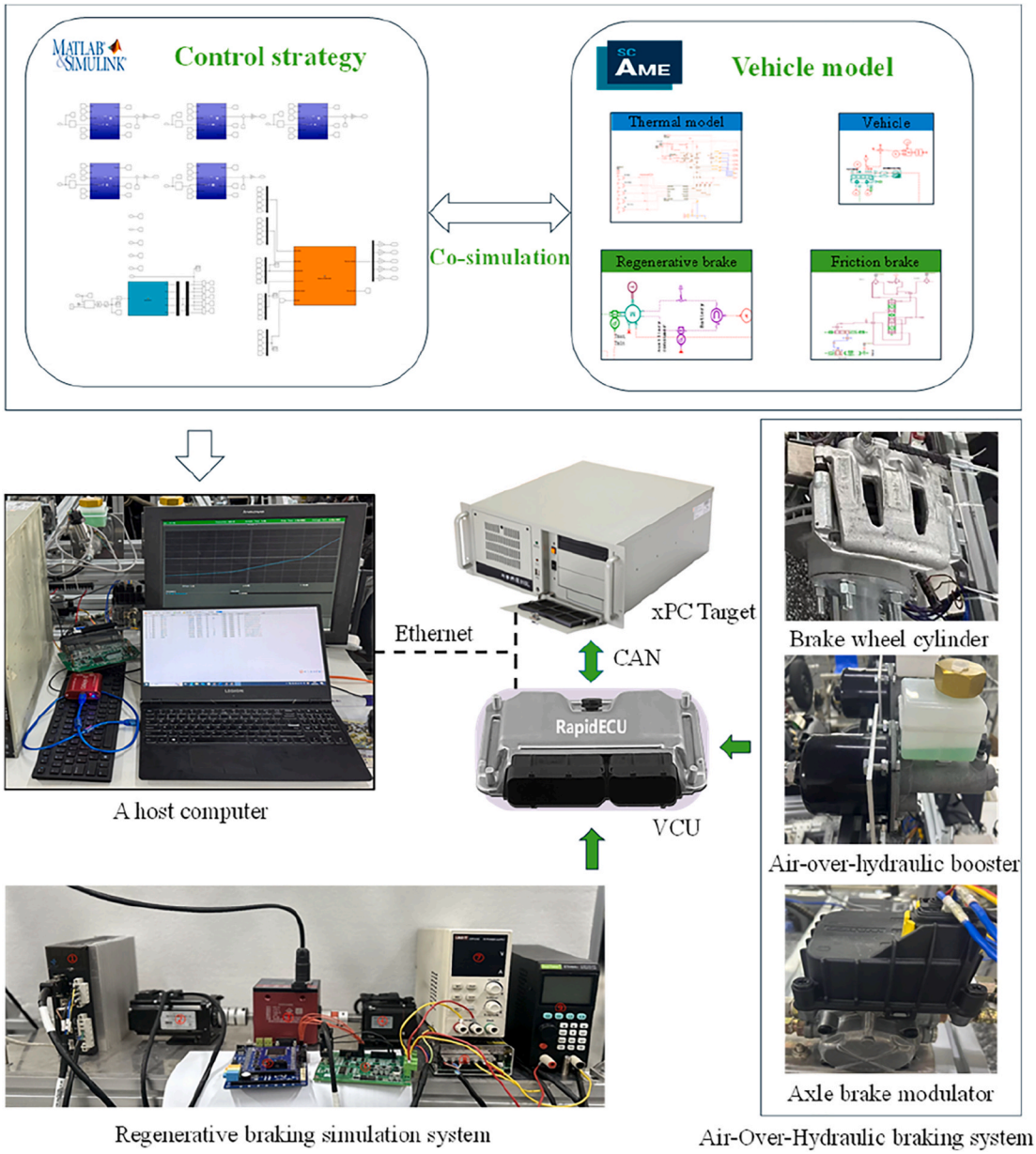


Fig. 16. Multi-axle vehicle composite braking system simulation platform.

capable of deterministic fixed-step computation, enabling millisecond-level real-time simulation of vehicle dynamics and providing high-frequency vehicle state parameters for VCU testing. The information of its processor and basic parameters is recorded in Table 3.

Following its modular design in MATLAB/Simulink, the compensation control algorithm is automatically converted into production-grade C code using the Embedded Coder tool chain. This code is then cross-compiled and deployed to the VCU, forming a real-time control system suitable for rapid control prototyping. The communication link between the VCU and the xPC Target system is established over a CAN 2.0B bus, providing a reliable, real-time channel for bidirectional data exchange. This includes control commands like target braking torque and vehicle status parameters such as wheel speeds and brake chamber pressures.

This HIL test platform accurately replicates the dynamic characteristics and network communication environment of an actual vehicle braking system, as well as high repeatability. It thereby serves as an effective basis for assessing the dynamic response, real-time performance, and robustness of the proposed compensation control strategy under near-realistic operating conditions.

This study constructs three typical conditions: friction braking compensation condition under motor priority braking, electromechanical synergistic compensation condition under partial motor overheating, and pure friction braking compensation condition under emergency braking. For the above working conditions, two groups of comparative simulation analysis of “without compensation” and “with compensation” are carried out to achieve quantitative evaluation and verification of the performance of the compensation control strategy.

4.2. Friction braking compensation working condition under motor priority braking

Fig. 17 shows the normal motor priority braking and the friction brake providing the residual braking torque. This operating condition is the mode in which the system operates under normal conditions with the aim of maximizing the energy recovery efficiency. In this scenario, the initial brake disc temperature is approximately 380 °C, the vehicle decelerates from 60 km/h to rest, and the braking intensity is 0.3g.

As illustrated in Fig. 17(a) and (b), when braking from 60 km/h, the vehicle employing the non-compensated strategy exhibits braking time and distance exceeding those of the compensated strategy by 0.071 s and 0.964 m, respectively. Analysis of Fig. 17(c) indicates that under non-compensated conditions, the friction braking torque on each axle decreases during the initial braking phase because of insufficient motor torque output. Combined with the motor torque characteristic curve in Fig. 17(d), the variation in motor torque before and after the application of compensation control is solely attributable to changes in vehicle speed. Since the motor’s compensation capacity has reached saturation, the compensation mechanism is not activated to provide additional torque adjustment. At this time, the system is incapable of leveraging the motor’s residual compensation capability to dynamically mitigate the degradation of frictional braking torque. Following implementation of the friction braking compensation strategy, as shown in Fig. 17(e), wheel cylinder pressure modulation maintains axle friction torque at the initial thermal fade threshold, thereby achieving braking torque dynamic equilibrium. Correspondingly, as shown in Fig. 17(f), the compensation strategy increases the wheel cylinder pressure, thereby enhancing the braking torque and consequently raising the brake disc temperature relative to the control condition without compensation.

Table 3
Basic parameters of vehicle controller processor.

Technical specifications	Description
Main Processor	S32K344, 32-bit, Multi-core, 160 MHz, Hardware Floating Point Unit, ISO26262
Memory	Flash 4 MB, SRAM 512 KB, EEPROM 64 Kb
CAN	6x CAN FD, CAN compatible, compliant with CAN2.0B, ISO11898
Program Flashing	Bootloader based on CAN bus
Functional Safety	ASIL D

4.3. Mechanical-electrical synergistic compensation condition under partial motor overheating

Fig. 18 shows the partial motor overheating condition. This condition assumes that after multiple braking, the motor on one axle is overheated or unusable. This axle uses pure friction braking, and other axles use motor braking first to ensure maximum energy recovery. This condition is designed to test the fault-tolerant control of the system when some components fail. The initial braking temperature of the brake disc is about 380 °C, and the vehicle speed is braked from 60 Km/h to 0, with a braking intensity of 0.3g.

Fig. 18(a) and (b) demonstrate that under this braking condition, the non-compensation strategy yields 0.220s longer stopping time and 1.957m longer braking distance compared to the compensation strategy, confirming enhanced vehicle braking efficiency. Analysis of Fig. 18(c) indicates that without the compensation strategy restricts torque output from the axle 1 motor, preventing braking torque generation. The friction braking torque on axle 1 will gradually decrease with the braking time, while the friction braking torque on the other axes will only drop temporarily at the beginning of braking due to insufficient motor braking torque. In contrast, the compensation strategy prioritizes the allocation of the motor’s remaining compensation capacity to maintain the friction braking torque at its initial attenuation threshold, before the motors on each axle become capable of compensation, thereby ensuring the dynamic stability of the total braking torque. Fig. 18(d) demonstrates that without compensation control, the motors on each axle are braked first, and the maximum torque that the motor can provide will gradually increase as the vehicle speed decreases. During compensation control, motors are sequentially activated based on the magnitude of their remaining compensation capacity. The motor on axle 2 dynamically compensates for friction braking attenuation on axle 1, thereby forming a closed-loop control of the braking torque. Fig. 18(e) shows wheel cylinder pressure variations across all axles during friction braking compensation control. Through modulated pressure elevation, axle friction torque attains specified target values. Correspondingly, Fig. 18(f) shows that the compensation strategy increases the wheel cylinder pressure, which enhances braking torque and consequently raises the brake disc temperature compared to the non-compensated control condition.

4.4. Pure friction braking compensation condition under emergency braking

Fig. 19 shows the performance of multi-axle vehicles and braking systems under emergency braking conditions. This condition is used to verify the functional safety guarantee of the system in extremely critical and safety-related situations. In this condition, the vehicle adopts pure friction braking, the initial braking temperature of the brake disc is about 380 °C, the vehicle speed is braked from 60 Km/h to 0, and the braking intensity is 0.7g.

Fig. 19(a) and (b) show that, following emergency braking, the braking time and distance without the compensation strategy are extended by 0.490 s and 3.069 m, respectively, compared to those with compensation. This indicates that the compensation strategy effectively enhances braking efficiency. Fig. 19(c) shows that, under the no compensation strategy, the braking torque on each axle gradually declines over time, whereas with the compensation strategy, it remains stable at the corresponding constant value. This phenomenon is explained in Fig. 19(d), which shows that without the compensation strategy, the brake wheel cylinder pressure gradually decreases over the braking period. This phenomenon primarily stems from pressure decay

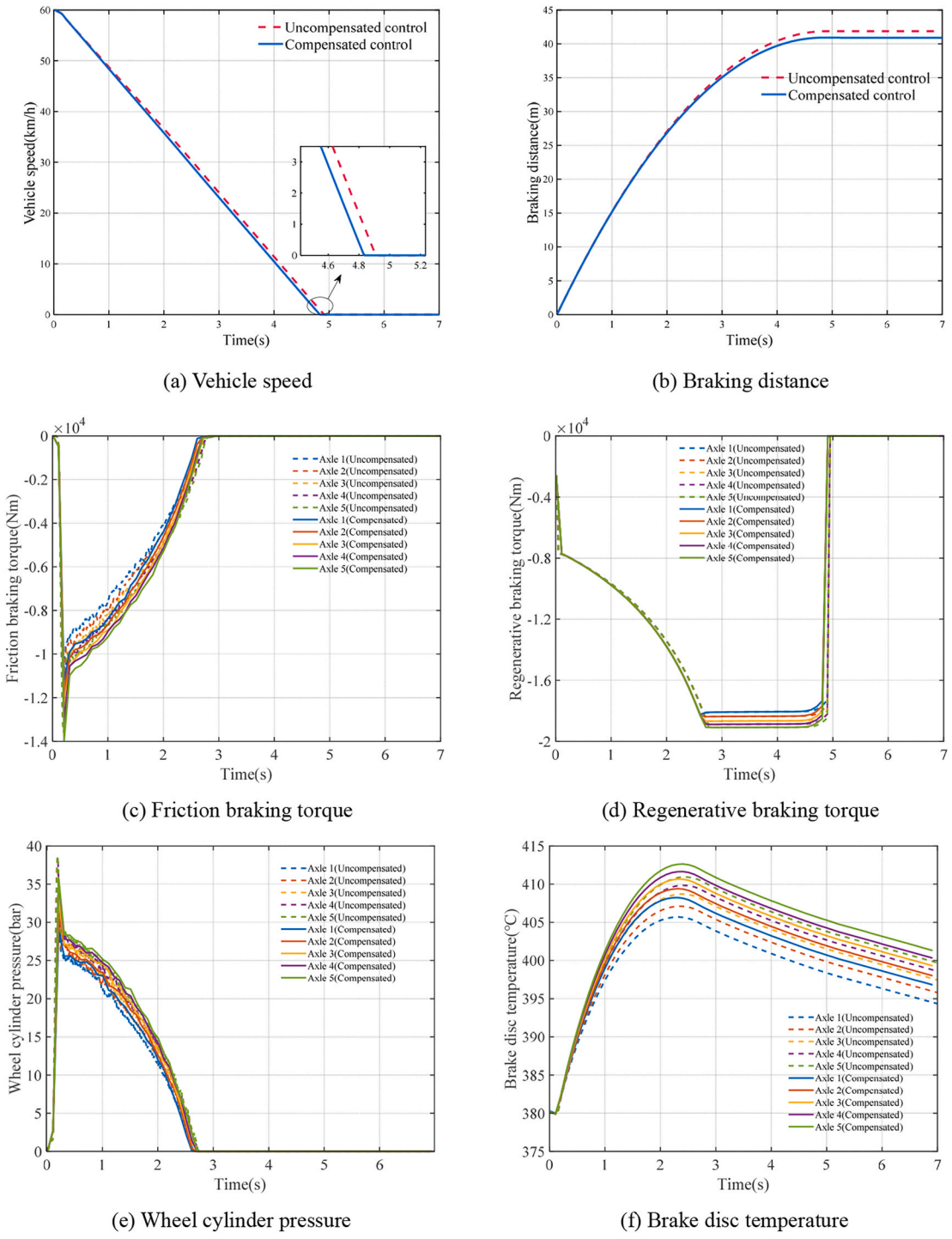
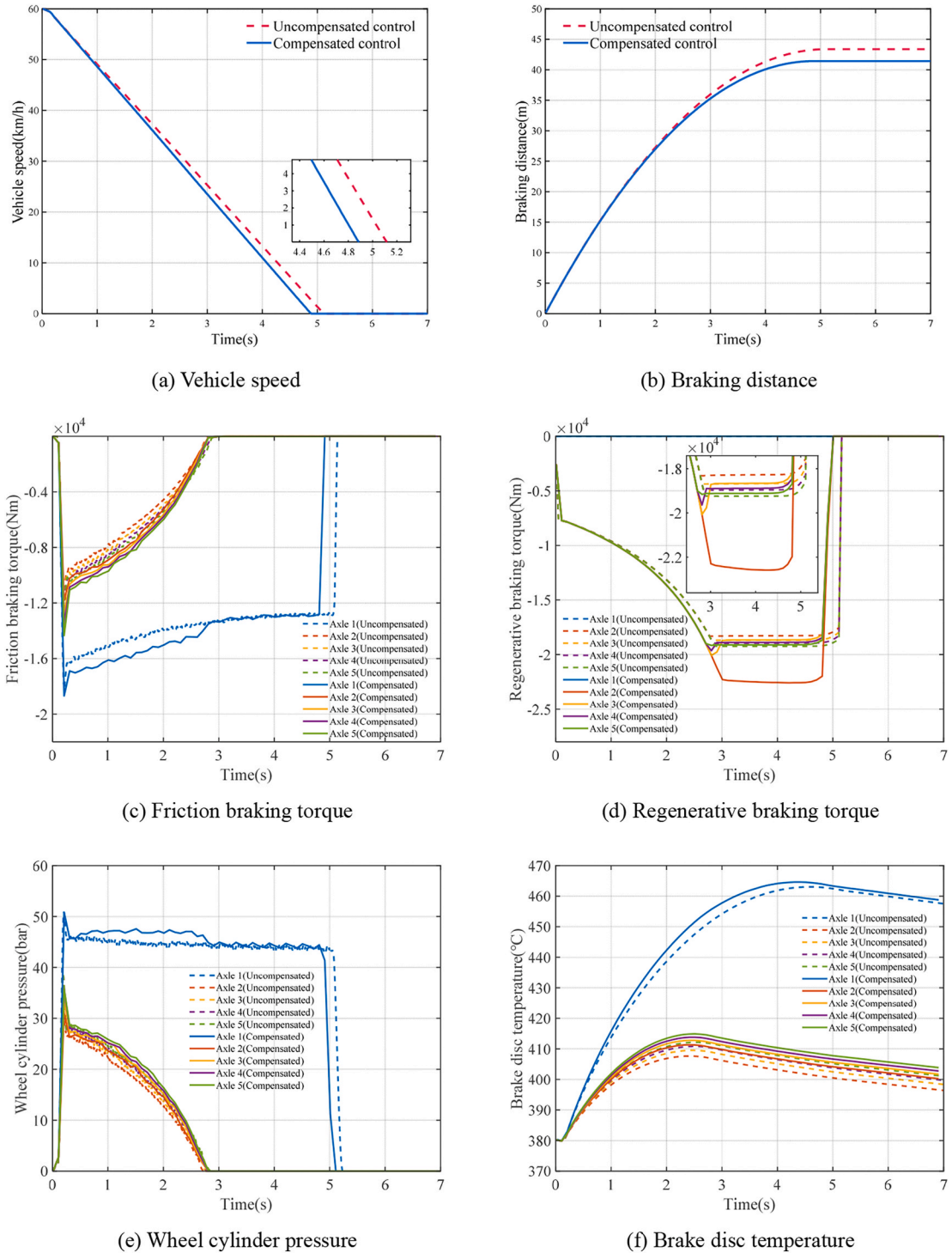


Fig. 17. Normal motor priority braking working condition.

in the pneumatic system's air reservoir. The compensation control strategy modulates wheel cylinder pressure to counteract this attenuation, sustaining axle-specific braking forces at predetermined setpoints, thereby providing a constant and large deceleration for the vehicle. Fig. 19(e) shows that, after adopting the compensation control strategy, the pressure loaded on the brake of each axle increases due to the increase in the braking force of each axle, which causes the temperature of the brake disc on each axle to be higher than that of the non-compensation strategy. However, under the compensation control, maintaining an adequate braking torque is the decisive factor for achieving the safety goal. Moreover, the closed-loop control architecture of this system has certain robustness and can compensate for such disturbances through real-time torque adjustment. Correspondingly, Fig. 19(f) shows that, after implementing the compensation strategy, the brake disc friction coefficient on each axle decreases more rapidly than in the non-compensation case, due to the faster temperature rise. Therefore, this pure friction braking compensation strategy can only achieve good results under short-time braking conditions. Otherwise, if the braking time is too long, the friction coefficient will decrease too much due to excessive temperature rise, which will hurt the braking performance.



(a) Vehicle speed

(b) Braking distance

(c) Friction braking torque

(d) Regenerative braking torque

(e) Wheel cylinder pressure

(f) Brake disc temperature

Fig. 18. Motor-limited condition.

5. Conclusion

This study investigates the composite braking system of a five-axle electric vehicle, focusing on braking force attenuation caused by motor overheating and thermal fade of the friction brake. A thermal analysis model of the composite braking system is established, and a control strategy based on dynamic axle load distribution and thermal fade characteristics is proposed. The strategy prioritizes motor braking compensation to mitigate braking performance degradation, thereby effectively improving both braking safety and energy efficiency.

(1) Based on the physical structure and working principle of the five-axis electric vehicle composite braking system, this paper establishes the regenerative braking system and air-over-hydraulic braking system models. Subsequently, the thermal analysis of the brake components and motor parts is conducted using the thermal network method, which enables real-time and accurate acquisition of the brake disc temperature and the motor winding temperature. Then, an experimental test is carried out on the brake disc to obtain the mapping relationship between temperature, friction coefficient and braking torque attenuation. Based

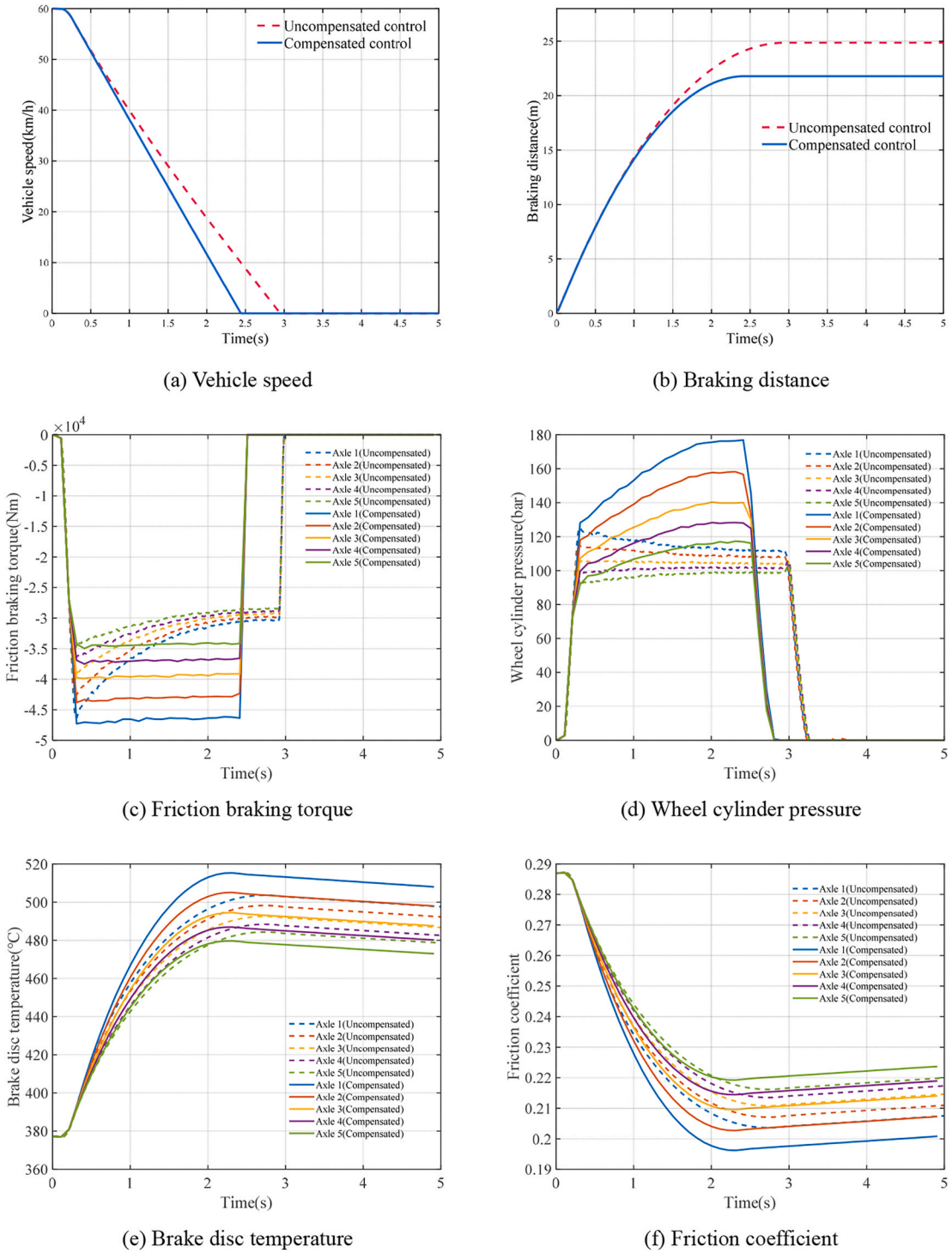


Fig. 19. Emergency braking condition.

on these findings, a new thermal fade compensation control strategy is proposed.

(2) To address the braking safety degradation induced by thermal fade, this study develops a real-time dynamic axle load distribution model for a five-axle vehicle during braking. To maximize energy recovery, a braking compensation strategy is proposed that prioritizes the use of the motor's residual capacity, with any remaining deficit compensated by friction braking. This integrated approach is validated through co-simulations using AMESim and Simulink under three distinct operating conditions, resulting in reduced braking times of 0.071 s, 0.220 s, and 0.490 s, respectively. Corresponding reductions in braking distance are 0.964 m, 1.512 m, and 3.069 m. These results confirm that the proposed control strategy effectively enhances vehicle safety.

CRediT authorship contribution statement

Shiwei Xu: Writing – review & editing, Funding acquisition, Conceptualization. **Wuzhi Wang:** Writing – original draft, Validation, Methodology, Conceptualization. **Xuebo Li:** Validation, Investigation.

Xuan Zhao: Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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